Paper Discussion

Group 8
Darin Biserta,
Ivanna Rubio Leiva,
Jan Smuts,
Maja Neumayer,
Zita Szekér

Kose, M. A., Prasad, E. S., & Terrones, M. E. (2009). Does financial globalization promote risk sharing? Journal of Development Economics, 89(2), 258-270.

Motivation & Research Questions

Background:

- In theory, financial globalisation → smoothing consumption growth across countries despite country-specific output fluctuations.
- Developing countries:
  - higher consumption volatility → higher theorized welfare gains from stabilization.
- However, despite increasing cross-border capital flows since the mid-1980s are welfare gains not completely realised for non-industrial nations.

Relation to IFM:

- Empirically testing theorized relationship of cross-border capital flows and RS & practical implementation of measuring financial integration

Research Questions:

- *What is the impact of financial globalisation on the degree and evolution of international consumption risk sharing for industrial and developing countries?*
- *Can the composition of capital flows (FDI, equity, debt) explain why some countries have failed to realise the presumed benefits of globalisation?*

Related literature:

- Limited degree of risk sharing even in advanced economies (Obstfeld 1994, 1995)
- Conflicting trends in recent periods of globalisation (Sorensen, 2007; Moser et al., 2004)

  → Research gap: Sparse prior work for non-industrial economies often using samples that do not cover the recent wave of financial globalisation

Data

Macroeconomic Data

- Real per capita GDP
- Real Private consumption
- Real Government consumption

Sourced from (Penn World Tables 6.2 and World Bank WDI)

Integration Measures

- ***De jure***: binary measures of capital account restrictions (IMF's AREAER)
  - Dates of equity market liberalization;
  - 2 other continuous measures (Chinn and Ito, 2006; Edwards 2005)
- ***De facto***: Gross stocks of external assets liabilities relative to GDP (External Wealth of Nations Database)

Key Country Groups (69 total)

- ***Industrial Countries (21)***: Advanced economies used as the benchmark for high integration
- ***Emerging Market Economies (21)***: Developing countries that have significantly opened their capital accounts and received large capital inflows
- ***Other Developing Countries (27)***: Less integrated nations with more restricted capital accounts

Sample Coverage

- Time Period: **1960–2004** (Annual data)
- Globalisation Period: Sub-sample of 1987–2004 to capture the modern era of trade and financial liberalisation

Methodology

The Basic Risk-Sharing Framework

- The paper utilises a **standard regression model** to evaluate the degree of consumption risk sharing

$$\Delta \log c_{it} - \Delta \log C_t = \text{constant} + \beta_t(\Delta \log y_{it} - \Delta \log Y_t) + \varepsilon_{it}$$

- $\Delta\log c_{i,t}$, $\Delta\log y_{i,t}$ → country-specific consumption and output growth
- $\Delta\log C_t$, $\Delta\log Y_t$ → represent common world components

Theoretical Benchmark

- Perfect risk sharing: □ = 0 (idiosyncratic consumption growth is decoupled from income shocks)
- No risk sharing: □ = 1 (consumption growth perfectly tracks output growth)

Empirical Approaches

1. ***Cross-Sectional Analysis***: estimate the regression for **each year**, tracing the evolution of risk sharing index 1-□, from 0 (none) to 1 (perfect)
2. ***Time-Series Analysis***: model for **each country** over rolling nine-year periods to capture country-specific changes over time
3. ***Panel Regressions***: Extended models used to evaluate the direct impact of financial globalisation (de jure & de facto) and the composition of capital flows on the risk-sharing coefficient

$$\Delta \log c_{it} - \Delta \log C_t = \text{constant} + \mu_t(\Delta \log y_{it} - \Delta \log Y_t) + \gamma_t' \text{FO}_{it}(\Delta \log y_{it} - \Delta \log Y_t) + \varepsilon_{it}$$

Main Results

1. **Risk sharing improved over time (modest increase in 1-□)**
   - Consumption volatility relative to output declined → indicates partial decoupling of domestic consumption from income shocks
   - Significant improvement after the late 1980s

2. **Advanced economies benefit more**
   - Industrial countries drove the global trend → index 1-□ rose from approx. 0.15 (1970s) to 0.50 (post-1987)
   - Higher consumption correlations (implies more efficient pooling of idiosyncratic risk)
   - Stronger financial systems + legal frameworks → lower transaction costs for international asset trade

---

3. **Emerging markets show weaker improvements despite high integration**
   - Capital flows often correlated with domestic cycles → inflows during booms, outflows during recessions → amplifies rather than smooths shocks
   - Risk sharing limited during financial crises→ □ coefficient approaches 1 (zero insurance) during "sudden stop" episodes

4. **Composition of capital flows matters**
   - Equity and FDI improve risk sharing → "Variable payouts" act as automatic stabilizers
   - Debt flows provide weaker insurance → fixed repayment obligations regardless of economic health → often destabilizing for developing nations

Contribution & Conclusion

Contribution to the empirical literature

- Extend risk-sharing analysis to large sample of emerging & developing markets
- Examine changes in risk sharing over time
  → relate to changes in the degree of financial openness
- Careful evaluation of multiple measures of risk sharing

Conclusion

- Financial globalization ***modestly*** improves risk sharing
  - Underwhelming compared to theorized effects
- Over the recent globalization period:
  - Benefits are stronger in advanced economies
  - limited to almost non-existent for emerging markets
- Domestic financial development and institutions are crucial
  - EMEs and Developing countries → more integrated into global financial markets

Major Comments & Suggestions

Strengths

- Rigorous evaluation of the relation between International Risk Sharing & Financial openness
- Critical, in depth, connection with prior literature

Potential limitations & Caveats

- ***Measurement error***
  - Consumption data in developing countries may be unreliable
- ***Endogeneity of financial openness***
  - Countries that liberalize capital markets may already have better institutions and macroeconomic stability
- ***Procyclical capital flows***
  - Capital inflows to emerging markets tend to rise during booms and reverse during crises →Limiting risk-sharing benefits

Endogeneity of De Jure Liberalization measure

- Capital account liberalization may be endogenous
- Countries with stronger domestic institutions, stable macroeconomic/political environments & deeper financial systems are both more likely to liberalize and more likely to exhibit higher consumption correlation **potential selection bias**

Possible methodological improvements:

- Instrumental variables
- Difference-in-differences around liberalization events

Minor Comments

Heterogeneity within country groups

- country groups (industrial, emerging, developing) may still be highly heterogeneous
- Additional robustness checks could test whether results change when certain countries are excluded/added

Statistical uncertainty in figures

- adding confidence intervals in Figures 1 and 2 would help evaluate the statistical significance of differences across country groups

Unclear policy implications in conclusion

Alternative measures of risk sharing

- The analysis focuses mainly on consumption-output correlations
- Using additional indicators (e.g., international asset income flows) could provide complementary evidence

References & Appendix

For the visuals of the "Main Results" presented see:

- Figure 1: Evolution of Risk Sharing Over Time
- Figure 2: Risk Sharing Across Country Groups
- Table 3: Panel Regression Results

Support for major comments 1:

- Lewis, K. (1996) What Can Explain the Apparent Lack of International Consumption Risk Sharing? https://www.nber.org/system/files/working_papers/w5203/w5203.pdf
- Sørensen & Yosha (1998) International Risk Sharing and European Monetary Union https://www.uh.edu/~bsorense/jiepublished.pdf

Support for major comments 2:

- Bekaert, Harvey & Lundblad (2005) – financial liberalization and growth https://www.sciencedirect.com/science/article/abs/pii/S0304405X04002193
- Kose, Prasad, Rogoff & Wei (2009) – financial globalization benefits depend on institutions https://prasad.dyson.cornell.edu/doc/nontechnical/KosePrasadRogoffWei.F&DMarch2007.pdf

Appendix 1

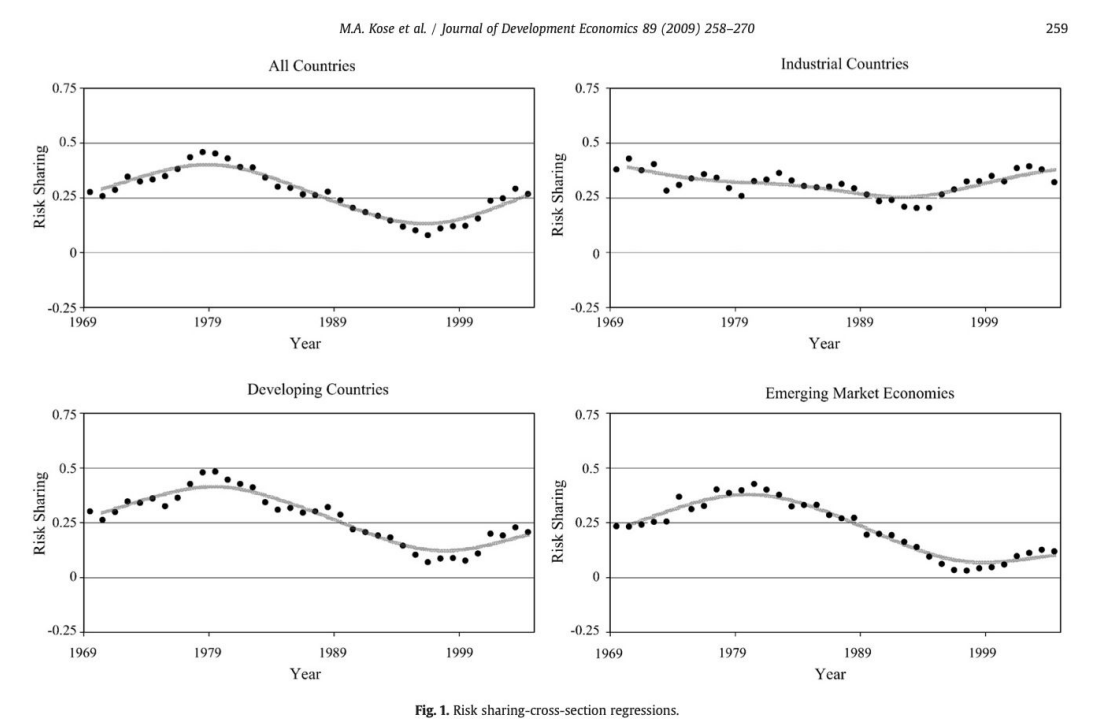

Fig. 1. Risk sharing-cross-section regressions.

Key Insights from Figure:

- Industrial countries: higher and relatively stable risk sharing over time
- Emerging markets: sharp decline in risk sharing during late-1990s financial crises
- Developing countries: consistently lower levels of risk sharing
- Overall: only modest improvement in global risk sharing in the early 2000s

Appendix 2

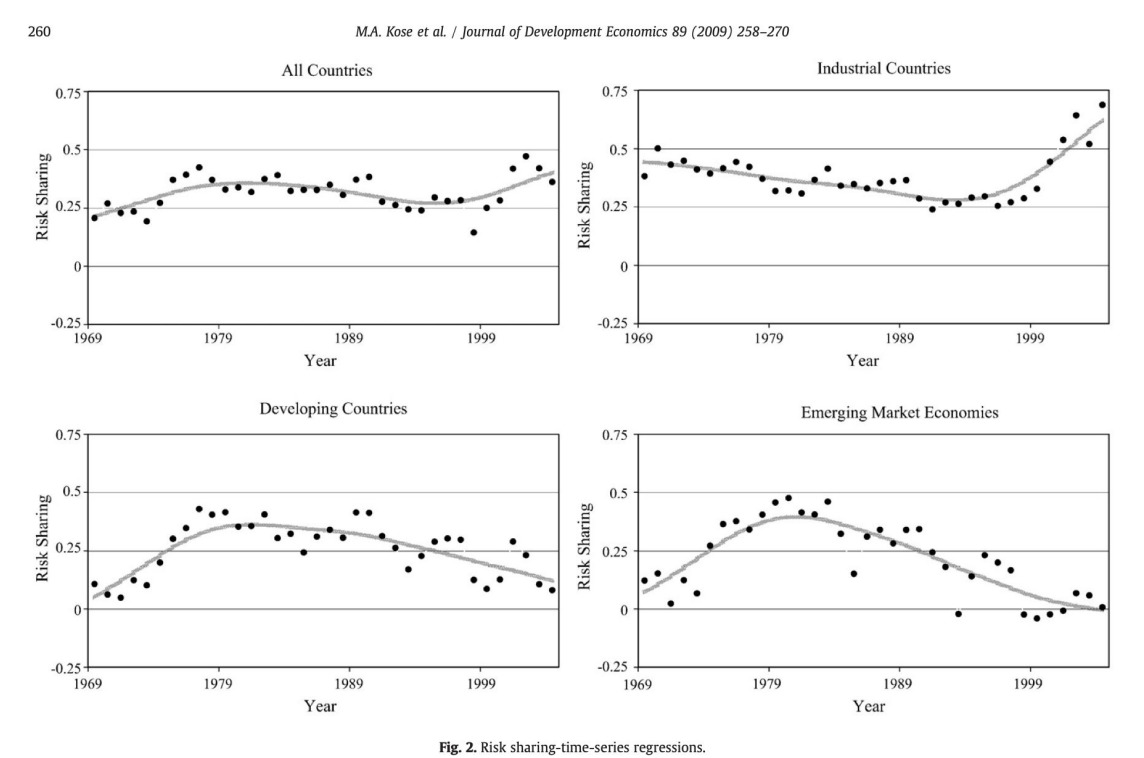

Fig. 2. Risk sharing-time-series regressions.

Key Takeaways:

- Higher and more stable risk sharing in industrial economies
- Sharp decline in emerging markets during late-1990s crises
- Limited improvements in developing countries
- Overall improvement remains modest

Appendix 3

Table 1b
Risk sharing and financial integration (de facto and de jure measures, globalization period, 1987–2004).

		De jure				De facto		De jure — IMF		De jure — Edwards	
		IMF	BHL	Chinn–Ito	Edwards	Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
I. All countries											
Output	0.860***	0.879***	0.848***	0.856***	0.924***	0.859***	0.928***	0.871***	0.936***	0.923***	0.980***
	[0.046]	[0.043]	[0.051]	[0.048]	[0.076]	[0.048]	[0.054]	[0.047]	[0.050]	[0.076]	[0.072]
Output × interaction[a]		−0.074	0.008	−0.028	−0.143			−0.086	−0.062	−0.146	−0.118
		[0.099]	[0.075]	[0.031]	[0.148]			[0.100]	[0.107]	[0.153]	[0.149]
Output × interaction[b]						0.003	−0.090*	0.027	−0.080	0.007	−0.083
						[0.043]	[0.047]	[0.045]	[0.052]	[0.034]	[0.057]
R2-adjusted	0.606	0.607	0.619	0.608	0.620	0.605	0.598	0.607	0.598	0.620	0.613
N	1193	1188	925	1184	924	1191	1185	1186	1180	924	920
II. Industrial countries											
Output	0.661***	0.721***	0.662***	0.713***	0.964***	0.704***	0.716***	0.783***	0.794***	1.088***	1.107***
	[0.026]	[0.073]	[0.030]	[0.090]	[0.284]	[0.029]	[0.031]	[0.081]	[0.081]	[0.296]	[0.300]
Output × interaction[a]		−0.064	…	−0.022	−0.341			−0.101	−0.099	−0.456	−0.474
		[0.075]	…	[0.042]	[0.317]			[0.073]	[0.074]	[0.328]	[0.334]
Output × interaction[b]						−0.037***	−0.041***	−0.028**	−0.033***	−0.019	−0.018
						[0.012]	[0.012]	[0.011]	[0.011]	[0.012]	[0.012]
R2-adjusted	0.617	0.625	0.643	0.624	0.643	0.618	0.617	0.626	0.626	0.643	0.643
N	357	352	277	348	277	359	359	354	354	279	279
III. Developing countries											
Output	0.873***	0.878***	0.836***	0.869***	0.890***	0.761***	0.979***	0.763***	0.979***	0.836***	1.030***
	[0.051]	[0.046]	[0.054]	[0.057]	[0.088]	[0.066]	[0.064]	[0.065]	[0.060]	[0.108]	[0.107]
Output × interaction[a]		−0.027	0.062	−0.010	−0.072			−0.087	−0.005	−0.104	−0.082
		[0.123]	[0.082]	[0.039]	[0.192]			[0.102]	[0.127]	[0.179]	[0.186]
Output × interaction[b]						0.312**	−0.120*	0.357**	−0.119*	0.229	−0.152**
						[0.132]	[0.060]	[0.137]	[0.062]	[0.185]	[0.067]
R2-adjusted	0.584	0.584	0.597	0.584	0.596	0.588	0.585	0.588	0.584	0.597	0.596
N	828	828	641	828	640	825	824	825	824	640	639
IV. Emerging market economies[c]											
Output	0.966***	0.984***	1.008***	0.938***	1.028***	0.890***	0.908***	0.895***	0.921***	0.987***	0.967***
	[0.048]	[0.050]	[0.045]	[0.047]	[0.058]	[0.069]	[0.094]	[0.068]	[0.094]	[0.086]	[0.125]
Output × interaction[a]		−0.136	−0.062	−0.059**	−0.119			−0.063	−0.141	−0.188	−0.084
		[0.093]	[0.077]	[0.025]	[0.114]			[0.116]	[0.100]	[0.120]	[0.130]
Output × interaction[b]						0.196	0.065	0.210	0.071	0.178	0.049
						[0.143]	[0.073]	[0.156]	[0.070]	[0.216]	[0.077]
R2-adjusted	0.797	0.799	0.813	0.800	0.812	0.804	0.797	0.804	0.799	0.816	0.812
N	357	357	276	357	275	356	357	356	357	273	275

Note: This table shows the results of panel regressions with yearly data. For details of the regression specification, see Section 4. The standard errors robust to heteroscedasticity and within-country serial correlation are in brackets. Regressions also include country fixed effects and year dummies. The symbols *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. "Assets" and "Liabilities" refer to gross foreign assets and gross foreign liabilities relative to GDP.

[a] De jure measure — either IMF, Bekaert–Harvey–Lundblad, Chinn–Ito or Edwards.

[b] De facto measure — either assets or liabilities.

[c] Emerging Market Economies are a part of the group of Developing Countries.

Key Insights

- Lower coefficients indicate greater international risk sharing
- Industrial economies exhibit stronger risk sharing
- Emerging markets show very limited risk sharing
- Overall improvements after financial globalization remain modest